Ian Burress

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EXPERIENCE

Application and Installation Engineer October 2025 - Present

Caterpillar | Houston, TX

- Led AI integration team across 27 pilot projects, automating manual tasks to save 2,500+ hours annually for an 18 person division
- Direct customer support for the 3500 Gas Generators and assistance in reintegration of 20 cylinder engine rated up to 10 MW
- Served as corporate liaison between global dealers and customers, facilitating effective communication, resolving technical inquiries, and ensuring alignment with Caterpillar's product support strategies

Engineering Rotational Development Program August 2024 - October 2025

Caterpillar | Houston,TX / Lafayette, IN /Clayton, NC

- Executed engine and chassis validation using Dewesoft (displacement sensors, thermocouples, pressure transducers, instrumented bolts, phototachs); delivered 3 test plans, and failure investigations utilizing DVP&R framework
- Rebuilt field-failure engine from block-up (360+ components, 17 kits, 28 discrepancies); led turbocharger FMEA and documented roading/bump tests for customer chassis across multiple operating points.
- Conducted warranty investigations and root cause analysis for marine, locomotive, oil & gas, and electric power systems; facilitated New Product Introduction (NPI) projects from warranty inception to engineering handoff.
- Led 6 in-field audits and comprehensive company/dealer-wide conference audit; resolved technical inquiries

Manufacturing Technology & Solutions Intern May 2023 - August 2023

Caterpillar | Chillicothe, IL

- Earned Six Sigma Green Belt leading DMAIC process to manage safety related improvement projects
- Developed an installation plan, for magnetic sensing of cabling for crane technologies, saving $250,000 if implemented

Graduate Research Assistant August 2022 - May 2024

Texas A&M University |College Station, TX

- Research in collaboration with Department of Energy and Dr. David Staack to complete a thesis design of a mobile radiation treatment facility for contaminated soil treatment
- Hosted weekly design reviews with research advisor to review project and strategize potential avenues of improvement
- Utilized SolidWorks to calculate facility weight and conducted Finite Element Analysis (FEA) to validate structural integrity

Integrated Components & Solutions Intern May 2022 - August 2022

Caterpillar | Peoria, IL

- Facilitated adoption of assisted reality into manufacturing environment. Activities included proof of concepts of Teamviewer product for work instructions, quality gates, gathered feedback to understand time usage and data inputs
- Oversaw production, contributed to fabrication troubleshooting with operators, provided feedback for enhanced end quality

Research Assistant November 2020 - May 2022

Michigan State University |East Lansing, MI

- Collaborated with Dr. Elisa Toulson to measure and record laminar flame characteristics of sequential two stage combustion
- Processed and evaluated images captured from experimental apparatus through a high speed camera operating a Schlieren Light source
- Leveraged MATLAB to gather data such as radius, area, and circularity to find laminar flame expansion speed

Manufacturing Quality Intern May 2021 - August 2021

Stellantis | Dundee, MI

- Initiated 2 CPIs, managed plant-wide audits, and evaluated 15 stations for defects deploying World Class Manufacturing strategies and databases such as EBOM and MQAS
- Produced 5+ Kaizens enhancing quality and saving over $50,000 annually in production and warranty costs

Interior Design Intern June 2020 - August 2020

Fiat Chrysler Automobiles | Auburn Hills, MI

- Hosted Q&A referencing knowledge of Autodesk, Photoshop, and various CAD softwares to build visualizations, collaborated cross functionally with 20+ other employees to ensure feasibility of an infotainment design concept
- Awarded most innovative concept, FCA to research idea for implementation and patent for future vehicles

Manufacturing Intern May 2019 - August 2019

Harley Davidson Motor Company | York, PA

- Collaborated to diagnose engine issues utilizing knowledge of assembly, maintenance, and performance
- Designed new parts, tools, and layouts through New Source Introduction to 2 facilities creating 5 different models

SKILLS

Software: Siemens NX, Autodesk, SolidWorks, Creo, PowerAutomate, PowerBI, Copilot Studio

Analysis: Thermal Desktop, Ansys , LS-DYNA, STAR-CCM+

Languages: Python, MATLAB, Microsoft Excel VBA

AWARDS

Personal: Texas A&M - Campus Sustainability Champion, B1G - Academic Success in Athletics Award

Team: Texas A&M Solar Car Racing Team - Team Perseverance Award, Texas A&M - 2025 National Commercial

Scholarships: Supporting Engineering Excellence, Army Disaster Relief Fund, Texas A&M Research Grant

Guest Speaker Appearance: 2023 The Solar Car Challenge Workshop, 2024 International Topical Meeting on Nuclear Applications of Accelerators, 2025 Innovators Education Foundation Conference

EXTRACURRICULARS

Zenith Series Race Engineer February 2026 - Present

Cambern Performance | Lansing, MI

- Race engineer for Ginetta G55 GTA in the ZR2 Class (maximum Power-to-Weight ratio of 9.0:1) across 6+ hour endurance races amongst a 30+ vehicle field, with a 5 man crew and 2 drivers
- Coordinating fuel stops and recording lap times as well as coordinating with mechanics for vehicle adjustments during stops
- In charge of maintaining crew, vehicle, and driver timeliness for track operations such as pregrid and post race inspection

Solar Car Dynamics and Body & Sizing Judge July 2025 - Present

Innovators Educational Foundation | Rolla, MO

- Race official for North American racing series hosting 50+ collegiate teams in a closed track and cross country endurance race
- Directly responsible for making sure all regulations regarding interior sizing and vehicle exterior meet track and road standards and qualifications

- Core group of race officials running pre race dynamic assessment qualification events including slalom, figure 8, wet brake, acceleration test, and u-turn test. Used these tests to assess road worthiness of student design vehicles
- Review and assist teams to ensure race ready compliance prior to event via design reviews and open lines of communication

Circuit of The Americas Fire Marshal March 2024 - Present
Circuit of The Americas | Austin, TX

- Served as Certified SCCA Pit Lane Intervention Marshal at major motorsports events (World Endurance Championships, NASCAR Cup Series, Mustang Challenge, Lamborghini Super Trofeo N.A.)
- Ensured safety for drivers, crew, and spectators by monitoring pit lane, manning fire extinguishers during refueling, and enforcing relevant safety protocols
- Required for fire suppression during fire during 2024 Ford Mustang Challenge, tire caught fire due to wheel lockout

FPV Drone Amateur Pilot March 2019 - Present

- Assembling FAA approved drone flying via first person point of view controls

Program Director & Founder July 2022 - August 2024
Texas A&M Solar Car Racing Team | College Station, TX

- Founded and led a design team with 14+ sub teams across 12+ majors, achieving a 7th place finish in North America
- Directed an 85+ person team through full life cycle of product development, resulting in a custom, solar-powered, road-legal electric vehicle to compete in The Formula Sun Grand Prix and American Solar Challenge
- Drove record-setting competition performance, including a top 5 fastest lap, and only first year program to qualify and finish cross country event since 2018. Team was recognized for team perseverance during The American Solar Challenge event

Business Director & Chassis Member September 2019 - May 2022
Michigan State Solar Car Racing Team | East Lansing, MI

- Directed business operations, financial transactions, and sponsor relations, managing an $125,000 dollar budget, raising over $30,000 in new sponsorships and managing competition logistics within 5% of actual cost
- Collaborated on a 12-person sub-team to reintegrate composite body panels with a redesigned chassis, establishing composite-to-metal mechanical fastings and bonded joints for optimal alignment and assembly tolerances.

EDUCATION

Texas A&M University | Master of Science in Mechanical Engineering May 2024

- M.S. Thesis Title: *"The Design of a Mobile Electron Beam Treatment Station For Contaminated Soils"*

Michigan State University | Bachelor of Science in Mechanical Engineering May 2022

- *NCAA Wrestler*; Michigan State;*Senior Business Director*, Solar Car Racing Team; *Research Assistant,* College of Engineering